

- (scraper) Modify the `get_songs` in 'songs.py' file to use the catalog instead of

scrapping again. (2 points)

I modified `get_songs()` so that it no longer scrapes data and relies solely on the catalog. Now the function loads `catalog.json` and directly iterates through the artists and songs already detected in `get_catalog()`. In `scraper/main.py`, I updated the `OUTPUT_DIRECTORY` and `LOGS_DIRECTORY` paths to match my project's structure.

```
def get_songs(output_directory: str, version: int = 0):

    catalog_path = Path(output_directory) / "catalog.json"

    try:
        catalog = files.load_from_json(catalog_path)
    except Exception as e:
        log.error(f"Error loading catalog from {catalog_path}: {e}")
        raise

    log.info("Starting lyrics download from catalog.json")
```

- (scraper) Check the logs for something strange. Try to know where those messages come from. No need to solve it, only get the origin. (0.5 points).

Looking at the logs, I see two messages that stand out. The first says that the file already exists, which simply means that the song was already downloaded and the program ignores it. The second is a repeated error regarding a variable `e`. This error stems from a piece of code that attempts to display an error that is actually poorly written.

```
13 11:29:08 INFO      Scrapper started at 2025-11-23 11:29:08.634678
13 11:29:08 INFO      Starting to download lyrics...
13 11:29:08 INFO      Starting lyrics download from catalog.json
13 11:29:08 INFO      Processing artist: Abel Pintos
13 11:29:08 INFO      File ./files/songs/abel_pintos/la_llave.txt already exists. Skipping download.
13 11:29:08 ERROR      Error downloading 'La Llave' from https://acordes.lacuerda.net/abel_pintos/la_llave
13 11:29:08 INFO      File ./files/songs/abel_pintos/sin_principio_ni_final.txt already exists. Skipping
13 11:29:08 ERROR      Error downloading 'Sin Principio Ni Final' from https://acordes.lacuerda.net/abel_p
13 11:29:08 INFO      File ./files/songs/abel_pintos/no_me_olvides.txt already exists. Skipping download.
13 11:29:08 ERROR      Error downloading 'No Me Olvides' from https://acordes.lacuerda.net/abel_pintos/no_
13 11:29:08 INFO      File ./files/songs/abel_pintos/bailando_con_tu_sombra.txt already exists. Skipping
13 11:29:08 ERROR      Error downloading 'Bailando Con Tu Sombra' from https://acordes.lacuerda.net/abel_p
13 11:29:08 INFO      File ./files/songs/abel_pintos/once_mil.txt already exists. Skipping download.
13 11:29:08 ERROR      Error downloading 'Once Mil' from https://acordes.lacuerda.net/abel_pintos/once_mil
13 11:29:08 INFO      File ./files/songs/abel_pintos/motivos.txt already exists. Skipping download.
13 11:29:08 ERROR      Error downloading 'Motivos' from https://acordes.lacuerda.net/abel_pintos/motivos.s
13 11:29:08 INFO      File ./files/songs/abel_pintos/bailando_con_tu_sombra_aleli.txt already exists. Ski
13 11:29:08 ERROR      Error downloading 'Bailando Con Tu Sombra Aleli' from https://acordes.lacuerda.net/
13 11:29:08 INFO      File ./files/songs/abel_pintos/el_avidino.txt already exists. Skipping download.
13 11:29:08 ERROR      Error downloading 'El Adivino' from https://acordes.lacuerda.net/abel_pintos/el_adi
13 11:29:08 INFO      File ./files/songs/abel_pintos/como_te_extranio.txt already exists. Skipping downlo
```

- **(cleaner)** The cleaner is getting the catalogs also as inputs. Although it is not a problem for now, we need to avoid it. Implement a solution. (0.5 points).

In the cleaner, it's necessary to prevent it from processing non-song files, because it was also cleaning the catalog and other project folders. I've made it so it only looks within /songs/. I also fixed a problem that caused duplicate folders to be created every time the cleaner ran, because the original code was accumulating old paths. I've adjusted the paths and added filters to exclude folders.

```
def list_files_recursive(path):  
    valid_files = []  
  
    for root, dirs, files in os.walk(path):  
        root_norm = root.replace("\\", "/").lower()  
  
        if "/cleaned" in root_norm:  
            continue  
        if "/validations" in root_norm:  
            continue  
        if "/lyrics" in root_norm:  
            continue  
        if "/results" in root_norm:  
            continue  
  
        if "/songs/" not in root_norm:  
            continue
```

- **(Validator)** There is an issue when reading/saving the files as it is creating more directories than needed. (0.5 points).

The original code copied the entire subfolder structure exactly as it came from the cleaner, even though we only needed to save the files as "OK" or "KO".

I solved this by having the validator save only the filename, without replicating any subfolders. This way, each validated song goes directly to ok/ or ko/, without duplicating folders or cluttering the project. In my case, since I don't have many songs, they all go to ko/.

```

filename = os.path.basename(file_path)

if validated:
    output_file = os.path.join(OUTPUT_DIRECTORY_OK, filename)
    OK += 1
else:
    output_file = os.path.join(OUTPUT_DIRECTORY_KO, filename)
    KO += 1

```

- **(Validator) Implement an additional validation rule (0.5 points)**

I've added two new rules to the validator: one that checks if the song contains any real chords, and another that ensures the file has at least one line with sufficient content.

```

def has_valid_chord(song):
    """Checks if the tab contains at least one valid chord."""
    chord_pattern = r'\b[A-G]#[b]?(\m|maj7|sus2|sus4|dim|7|9)?\b'
    return re.search(chord_pattern, song) is not None

def has_meaningful_line(song):
    """Returns True if there is a line with more than 10 characters."""
    for line in song.splitlines():
        if len(line.strip()) > 10:
            return True
    return False

```

- **Propose any change to make the code cleaner, clearer and better. (0.5 points).**

Better route organization:

A possible improvement would have been to organize all routes in one place, instead of having them scattered throughout the code. This would make it easier to make changes and avoid silly errors if the folder structure changes.

Improved process messages:

The messages displayed by the validator while it's working could also be improved to better explain why a file passes or fails validations. This would make it much easier to understand what's happening without having to review the code.